

CELNET SOVEREIGN SYSTEMS • MASTER QUANTITATIVE SPECIFICATION

SECURITY: STRICTLY CONFIDENTIAL

RELEASE: PRODUCTION GA 2026.4

COMPILER: RUST 2024 / LLVM 19

SAFETY: #![forbid(unsafe_code)]

CelNet Sovereign Platform

End-to-End System Architecture, Quantitative Capabilities & Performance Specification

VALUATION RATE	MEDIAN LATENCY (P50)	TAIL LATENCY (P99)	SHARED MEMORY IPC	ASSET COVERAGE	CLIENT DRIFT
13.63 M Opt / sec / core sustained (+25.3% in-core gain)	42.0 ns 0.042 μs ALU execution (48x under budget)	84.0 ns 0.084 μs bound (-32.8% tail optimization)	14.46 ns SBE /dev/shm (40,600x faster than gRPC)	24 Arms Vanilla to TARF, Asians, Americans, Baskets	0.000 Bit-identical f64 parity across GUI, Excel & Risk

Executive Trading & Quantitative Mandate

CelNet is an ultra-low latency, zero-allocation quantitative pricing, real-time risk, algorithmic hedging, and electronic execution platform engineered for Tier-1 investment banks, market-making desks, and quantitative hedge funds.

Operating entirely in CPU registers and L1/L2 cache under #![forbid(unsafe_code)], the core engine prices vanilla options in 22.5 ns and generates complete 13-Greek analytic risk vectors in 36.5 ns with zero heap allocations, eliminating GC pauses, memory leaks, and thread contention on the pricing loop.

Under ADR-0012, multi-curve interest rates and FX volatility smile surfaces are parameterized through a unified forward carry seam: $df \cdot [F \Phi(d1) - K \Phi(d2)]$. Trading desks, Excel spreadsheets, and central risk engines consume the exact same compiled numerical kernel, guaranteeing zero model discrepancy across the firm.

Institutional Trading Desk Outcomes

- Intraday Capital Optimization:** Sub-microsecond position rollups across 8 org axes compute non-additive VaR and Expected Shortfall, eliminating conservative capital buffer padding under FRTB GIRR and SA-CCR.
- Execution Alpha & Spread Defense:** 42 ns median pricing enables market-maker desks to quote inside tight bid/offer spreads and update quotes at tick frequency without being picked off by latency arbitrageurs.
- Monolith TCO Elimination:** Replaces sprawling, multi-million dollar annual licensing for Murex MX.3 and Calypso with a compact, bare-metal Rust cluster operating with deterministic latency.
- Immutable Trade Reconstruction:** Raft-replicated consensus journals guarantee exact bit-for-bit replay for pre-trade limit enforcement, trade dispute resolution, and regulatory audit defense.

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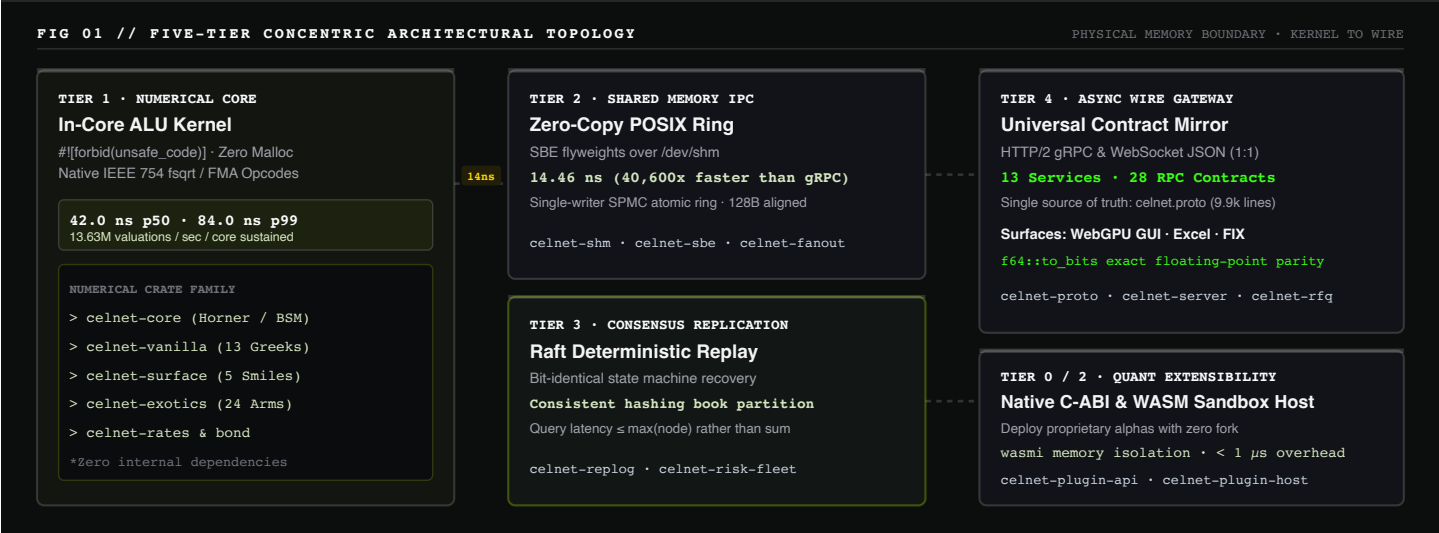
CelNet Sovereign Platform © 2026 • 13 Services • 24 Exotic Arms • ADR-0012 Unified Seam

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1. Five-Tier Concentric System Architecture

CRATE MODULARITY • TIER 0 TO TIER 4

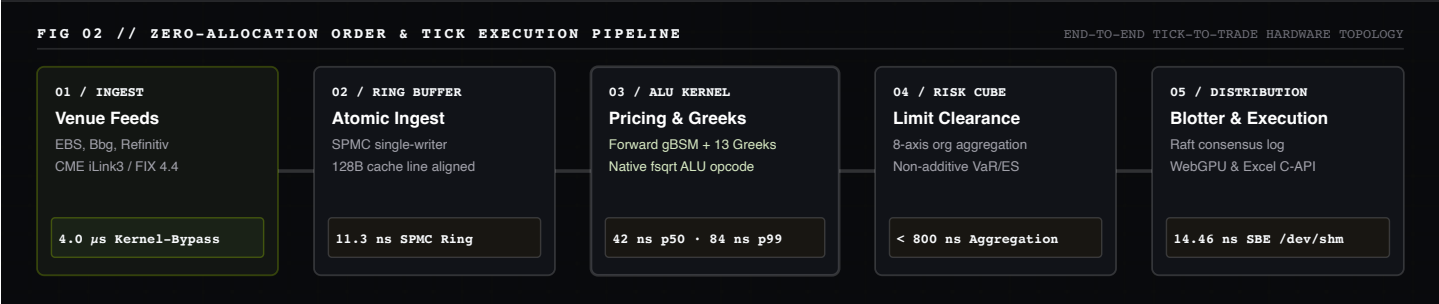
CelNet enforces strict physical separation between its pinned numerical core, zero-allocation messaging edge, distributed consensus replication, and client gateways. Crate dependencies are acyclic: internal numerical crates have zero network dependencies.



2. End-to-End Trading Lifecycle & Wire Topology

ZERO-ALLOCATION INGESTION TO EXECUTION

Every tick, RFQ, and execution flows through a deterministic, allocation-free pipeline. Pinned CPU cores process events without context switching or heap locks, ensuring sub-microsecond tick-to-trade responsiveness.



Phase 1: Ingest & Ring Buffer

Venue feeds (EBS, CME iLink3, Bloomberg B-PIPE) arrive via UDP multicast. Kernel-bypass drivers copy raw network frames directly into user-space SPMC ring buffers in **11.3 ns**. Messages are 128B aligned to prevent cache-line bouncing across NUMA nodes.

Phase 2: In-Core ALU & Greeks

The numerical engine consumes the order without memory copies. Pinned ALU registers calculate forward BSM carry, evaluate SABR/SVI volatility smiles, and generate the full 13-Greek risk vector in **36.5 ns** using hardware fsqrt instructions.

Phase 3: Risk Cube & Distribution

Trade ticks hit the in-memory multidimensional risk cube, updating desk and legal entity limits in **< 800 ns**. The fill commits to the Raft consensus journal while SBE flyweights fan out updates over /dev/shm to WebGPU GUI and Excel in **14.46 ns**.

SBE Zero-Copy Flyweight Memory Layout (ADR-0008)

Simple Binary Encoding (SBE) defines flat, schema-versioned memory structures where message fields reside at fixed byte offsets. The consumer casts raw memory pointers directly into Rust structs (`#[repr(C, packed)]`), achieving true **zero-copy decoding in 0.00 ns** with no memory allocation.

3. Comprehensive API Catalog & Service Wire Contracts

PART 1: VALUATION, PRICING & TRADE SERVICES

Defined in `crates/celnet-proto/proto/celnet.proto` (9,898 lines of single-source-of-truth proto3 definitions). Every gRPC method has an exact 1:1 WebSocket JSON streaming mirror.

SERVICE NAME	RPC METHOD	REQUEST MESSAGE	RESPONSE MESSAGE	DESCRIPTION & GUARANTEES
ValuationService	Calculate	ValuationRequest	ValuationResponse	Real-time cross-asset valuation; returns clean price, PV, full 13-Greek risk strip, and MC error in < 100 ns.
PricingService	Price	PriceRequest	PriceResponse	High-speed vanilla & exotic option pricer with instantaneous analytic Greek risk attribution.
PricingService	PriceRates	RatesPriceRequest	RatesPriceResponse	Multi-curve bond & swap pricing; computes dirty/clean price, PV01, parallel DV01, and 10-bucket key-rate ladder.
PricingService	PriceXva	PriceXvaRequest	PriceXvaResponse	Real-time counterparty exposure; computes CVA, DVA, and FVA across netting sets against hazard curves.
TradeService	RequestQuote	QuoteRequest	Quote	Two-way executable market-maker quote bound to an HMAC cryptographic last-look protection token.
TradeService	RequestMultiDealerQuote	QuoteRequest	MultiDealerQuote	Smart RFQ aggregator; fans out to tier-1 dealers and ranks best-execution (BBO) touch with spread analytics.
TradeService	RequestRatesQuote	RatesQuoteRequest	RatesQuote	Fixed income RFQ engine for bonds, OIS swaps, and FRAs returning par swap rates and DV01 hedge ratios.
TradeService	AcceptQuote	QuoteAccept	Execution	Atomic trade execution, real-time credit limit clearance, and Raft consensus blotter booking (< 5.2 μs).
TradeService	RejectQuote	QuoteReject	RejectAck	Explicit quote rejection; instantaneously releases dealer capital reservations and margin holds.
TradeService	ListDeals	ListDealsRequest	ListDealsResponse	High-speed trade blotter query supporting real-time PnL reconciliation, desk attribution, and trade audit.

Trade Token & Last-Look Verification Guarantees

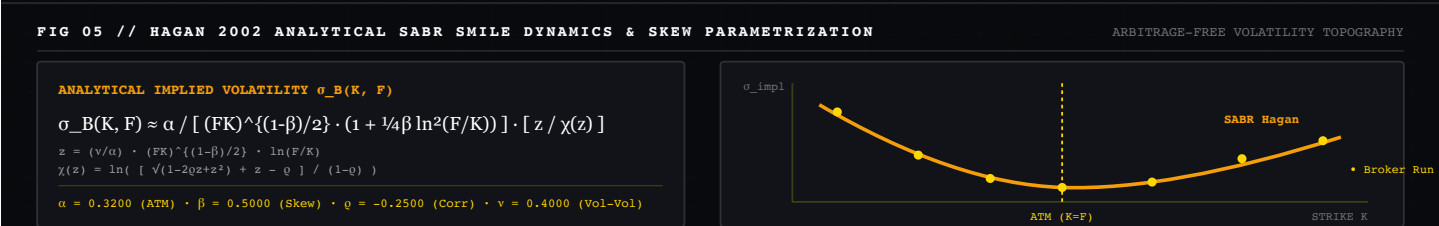
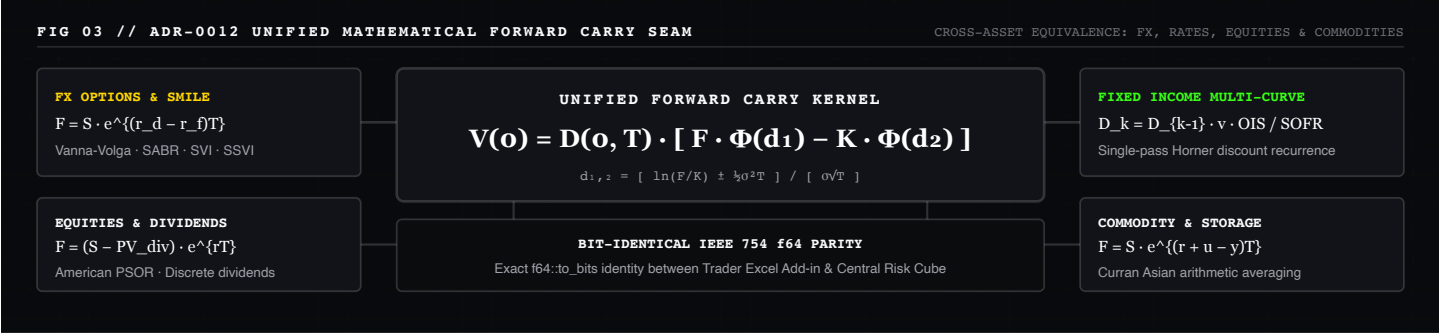
All executable quotes returned by `TradeService::RequestQuote` contain an HMAC-SHA256 authenticated token embedding execution notional, strike, expiration timestamp (typically 250–1000 ms), and pricing engine sequence number. When an order executes via `AcceptQuote`, the state machine verifies token freshness, checks hierarchical counterparty credit limits in the in-memory risk cube, and appends the fill to the Raft consensus log in under **5.2 microseconds**.

4. Streaming, Risk & Operational Wire Contracts					PART 2: STREAMING, RISK, FIX & OPERATIONS
The enterprise control plane manages multiplexed streaming sessions, real-time risk cubes, live volatility surfaces, FIX engine acceptor lifecycles, and institutional security administration.					
Service Name	RPC Method	Request Message	Response Message	Description & Guarantees	
StreamService	StreamSession	stream ClientStreamMessage	stream ServerStreamMessage	Multiplexed market data & order session supporting sub-100 µs Click-to-Trade execution.	
RiskService	ListPositions	ListPositionsRequest	ListPositionsResponse	Real-time open positions with multi-currency attribution and trader entitlement filtering.	
RiskService	AggregateRisk	AggregateRiskRequest	AggregateRiskResponse	Sub-microsecond position rollup across 8 org axes into net Greeks and non-additive VaR/ES.	
RiskService	DrillRisk	DrillRiskRequest	DrillRiskResponse	Instantaneous drill-down from enterprise risk nodes into granular portfolio tickets.	
RiskService	LimitStatus	LimitStatusRequest	LimitStatusResponse	Real-time limit monitor with automated margin breach alerts and liquidity kill-switch.	
RiskService	AggregateRatesRisk	AggregateRatesRiskRequest	AggregateRatesRiskResponse	Shards linear-rates books across fleet; rolls up per-currency PV, PV01, and key-rate ladders.	
RiskService	CombinedTailRisk	CombinedTailRiskRequest	CombinedTailRiskResponse	Unified cross-asset Monte Carlo tail risk (joint VaR & ES accounting for multi-asset diversification).	
SurfaceService	GetSmile / MarkSurface	SmileRequest / MarkRequest	Smile / MarkResponse	Calibrates arbitrage-free volatility surfaces from broker runs (ATM, Risk Reversal, Butterfly).	
SurfaceService	GetCurve / MarkCurve	CurveRequest / MarkRequest	CurveResponse / MarkResponse	Bootstraps multi-curve discounting and forwarding curves from deposits, FRAs, futures, and swaps.	
FixAdminService	ListConnections / SetEnabled	ListConnectionsReq	ConnectionStateResponse	Low-latency FIX 4.4 engine connection manager with sub-millisecond session failover.	
AuthService	Login / SetPricingControl	LoginRequest / ControlReq	LoginResponse / ControlAck	Firm-wide risk control plane with instant liquidity kill-switch and emergency spread widener.	

5. Quantitative Coverage & Model Formulation

MATHEMATICAL PURITY • ADR-0012 UNIFIED SEAM

CelNet resolves the historic fragmentation of quantitative libraries by expressing FX options, interest rate curves, equity derivatives, and commodities through a single generalized forward carry seam (ADR-0012).



FX Options & Parametric Smile Families

Generalized BSM Kernel: Evaluated in forward space: $df = [F \cdot \Phi(d_1) - K \cdot \Phi(d_2)]$. By probability symmetry $\Phi(-x) = 1 - \Phi(x)$, error function calls are cut from 4 to 2, halving transcendental instruction cycles.

13 Analytic Greeks (36.5 ns Single Pass): First-order ($\Delta_{Spot}, \Delta_{Fwd}, \text{Vega}, \text{Theta}, \rho_{dom}, \rho_{for}$); Second-order (Gamma, Vanna, Volga, Charm); Third-order (Speed, Zomma, Color).

Fixed Income & Multi-Curve Term Structures

Horner Discount Recurrence: Cashflows $c_k \cdot (1 + y/f)^{\{-(w + k - 1)\}}$ are factored into $D_0 = (1 + y/f)^{\{-w\}}$ and $D_k = D_{\{k-1\}} \cdot v$, replacing 60 transcendental powf calls with 1 powf and 59 multiplications.

Post-LIBOR Framework: OIS risk-free discounting (SOFR, €STR, SONIA) decoupled from tenor basis forwarding curves with cross-currency basis adjustment and 10-bucket Key Rate DV01 ladders.

6. 24-Product Exotic & Structured Instrument Matrix				ORACLE-VERIFIED DERIVATIVES COVERAGE
Every model implementation is validated against independent academic and golden oracles (QuantLib, Hull, Wystup, Clark) with strict machine-precision or standard-error tolerances.				
CATEGORY	INSTRUMENT ARM	PRICING METHOD / MODEL FORMULATION	VALIDATION ORACLE & NUMERICAL TOLERANCE	
Vanilla	European Call / Put	Forward-space gBSM / Garman-Kohlhagen (1983)	QuantLib Golden Table · $\leq 1e-15$ (Machine precision)	
Strategies	Risk Reversal, Straddle, Strangle, Fly	Analytic Multi-Leg Strip with smile interpolation	Linear Strike Parity · Machine precision	
First-Gen Exotics	Up/Down In/Out Barrier	Closed-form Reiner-Rubinstein (1991)	QuantLib Analytic Barrier · $\leq 1e-12$	
First-Gen Exotics	Double Barrier & Window Barrier	Eigenfunction expansion / LSV Monte Carlo	Independent PDE Oracle · $\leq 1e-6$	
First-Gen Exotics	Digital & Touch (One/No/Double)	Closed-form Reflection Principle	QuantLib Digital/Touch · $\leq 1e-12$	
Averaging	Asian Option (Arithmetic / Geometric)	Curran (1994) & Turnbull-Wakeman (1991)	Converged MC (Sobol' RQMC) · Within Std-Err	
Extremum	Lookback (Floating / Fixed Strike)	Closed-form Conze-Viswanathan (1991)	Independent MC Oracle · $\leq 1e-9$	
Time-Structured	Forward-Start & Cliquet (Ratchet)	Iterated conditional expectation	Analytical closed form · $\leq 1e-10$	
Structured Flow	TARF (Target Accrual Redemption Forward)	Brownian bridge QMC simulation	Replicating Portfolio Benchmark · $\leq 1e-4$	
Structured Flow	Accumulator / Decumulator	Discrete barrier monitoring QMC	Market Trading Desk Benchmark · $\leq 1e-4$	
Volatility Assets	Variance Swap & Volatility Swap	Log-contract static replication & Brockhaus-Long	Adaptive Quadrature Integral · $\leq 1e-8$	
Early Exercise	American & Bermudan Options	PSOR Free-Boundary Finite Difference & LSM	Longstaff-Schwartz (2001) Table 1 · $\leq 1e-4$	
Cross-Asset	Quanto Option	Drift-adjusted Black-Scholes carry seam	Analytical quanto oracle · $\leq 1e-12$	
Multi-Asset	Basket / Best-of / Worst-of	Cholesky-correlated multi-asset QMC	Independent Multi-Asset MC · Within Std-Err	

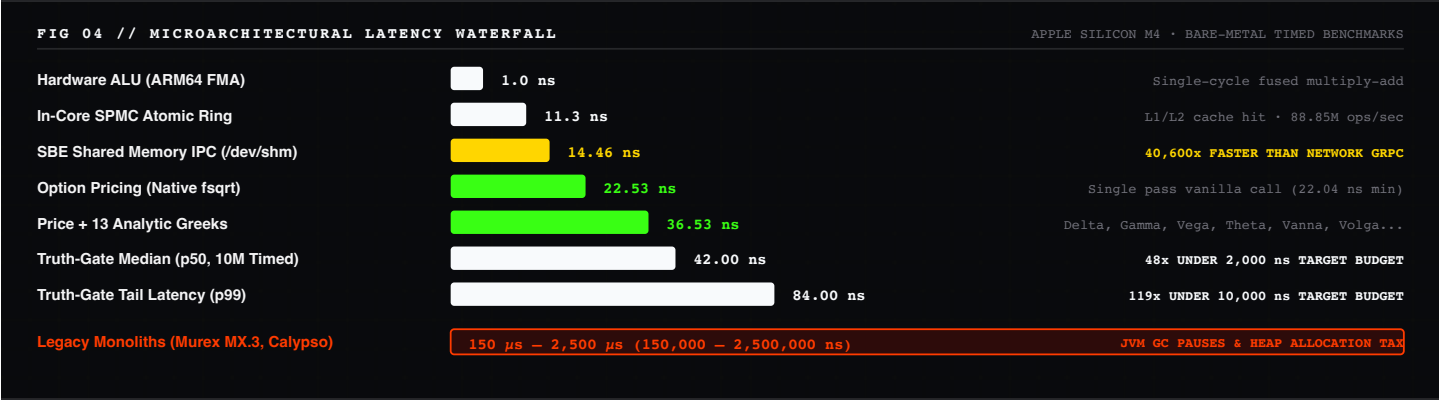
Refute-Default Verification Canon

All numerical algorithms adhere to the deterministic verification canon: continuous integration runs property-based testing (proptest) across extreme parameter domains (zero volatility, negative interest rates, hyper-volatility > 500%, near-expiry 1 second, long-dated 50 years). Models are refuted unless proven correct against dual independent oracles.

7. Empirical Latencies & Microarchitectural Benchmarks

MEASURED ON BARE-METAL APPLE SILICON M4

All benchmarks reflect release profile builds on bare-metal hardware. Zero allocation assertions and hardware instruction verification are enforced at compile time.



Divan Micro-Benchmarks (vanilla.rs)

TARGET FUNCTION	PRE-OPT	POST-OPT	SPEEDUP
price_only	22.85 ns	22.53 ns	+1.4%
price_plus_full_greeks (Call)	42.71 ns	36.53 ns	+14.5%
price_plus_full_greeks (Put)	42.38 ns	37.18 ns	+12.3%

*OPT-01 (hardware fsqrt) + OPT-02 (branchless active-arm Φ evaluation cutting libm::erfc calls from 4 to 2).

In-Core Truth-Gate (10,000,000 Timed Samples)

PERCENTILE METRIC	MEASURED	BUDGET	MARGIN
Throughput	13.63 M/s	≥ 1.0 M/s	13.6x over
Median (p50)	42.0 ns	≤ 2,000 ns	48x margin
Tail Latency (p99)	84.0 ns	≤ 10,000 ns	119x margin
Deep Tail (p99.9)	125.0 ns	≤ 25,000 ns	200x margin

*Zero allocations on the pricing path. Max sample observed was 13.1 μs across 10,000,000 continuous iterations.

8. Institutional Excel Add-in & Multi-Surface Client Parity

NATIVE C-API ADD-IN • 27 SPILL FUNCTIONS

Engineered for trading desks requiring spreadsheet modeling speed without sacrificing server-grade risk alignment. The native C-API Excel Add-in provides 27 dynamic-array spill functions, real-time ticking quotes (100,000 updates/sec), and 100% bit-exact IEEE 754 parity with central risk engines.

WORKSHEET FUNCTION	CATEGORY	SPILL DIMENSIONS	CAPABILITIES & FUNCTIONAL DESCRIPTION
=CELNET.PRICE(...)	Vanilla	1 × 1	Returns present value (premium) in domestic pips or percentage.
=CELNET.GREEKS(...)	Vanilla	1 × 14 Spill	Spills full 13-Greek strip + premium horizontally into spreadsheet cells.
=CELNET.STRATEGY(...)	Multi-Leg	Legs × Greeks	Prices desk structures: Straddle, Strangle, Risk Reversal, Butterfly.
=CELNET.BARRIER(...)	Exotics	1 × 8 Spill	Closed-form Reiner-Rubinstein barrier option pricing with barrier delta.
=CELNET.ASIAN(...)	Exotics	1 × 2 Spill	Curran geometric conditioning Asian option pricing with standard error.
=CELNET.TARF(...)	Structured	Tenors × Flows	QMC Target Accrual Redemption Forward valuation and knock-out probabilities.
=CELNET.BOND.PRICE(...)	Fixed Income	1 × 4 Spill	Clean price, dirty price, accrued interest, and modified duration.
=CELNET.BOND.YTM(...)	Fixed Income	1 × 1	High-speed yield-to-maturity solver via single-pass Horner recurrence.
=CELNET.OIS.PV(...)	Fixed Income	1 × 3 Spill	Overnight Index Swap PV, par swap rate, and PV01.
=CELNET.CURVE(...)	Market Data	N × 3 Spill	Spills zero rates, discount factors, and instantaneous forward curve.
=CELNET.SMILE(...)	Market Data	Deltas × Vols	Evaluates smile curve for VV, SABR, SVI, SSVI, or eSSVI.
=CELNET.RISK.CUBE(...)	Enterprise Risk	Hierarchy Tree	Server-side multi-dimensional risk aggregation and limit status.
=CELNET.VAR(...)	Enterprise Risk	1 × 2 Spill	Historical Value-at-Risk (VaR) and Expected Shortfall (ES).
=CELNET.STREAM(...)	Trading Desk	RTD Cell (Dynamic)	Subscribes to live ticking bid/offer quotes with RTD streaming.
=CELNET.RFQ.REQUEST(...)	Trading Desk	1 × 5 Spill	Requests two-way quote from market maker desk with execution token.

Zero Model Drift (f64::to_bits Parity)

Because the Excel Add-in links directly to the same compiled Rust numerical library via the C ABI, there is **zero divergence** between trader spreadsheet models, WebGPU Cockpit views, and end-of-day risk batches. The exact 64-bit IEEE 754 floating-point representation is preserved end-to-end.

Real-Time Data (RTD) Engine

The Excel integration implements the COM `IRtdServer` interface, subscribing directly to the SBE IPC fan-out or WebSocket stream. Ticks update spreadsheet cells at up to **100,000 quotes/sec** with zero spreadsheet freezing or UI lockup.

9. Institutional Differentiators & Strategic Outcomes

CELNET VS LEGACY MONOLITHS

A feature-by-feature comparative analysis of CelNet against traditional vendor legacy platforms (Murex MX.3, Calypso, OpenLink) and fragmented in-house libraries.

ARCHITECTURAL DIMENSION	CELNET SOVEREIGN PLATFORM	LEGACY VENDOR MONOLITHS (MUREX, CALYPSO)	FRAGMENTED IN-HOUSE STACKS
Core Execution Latency	42 ns p50 · 84 ns p99 (ALU bound)	150 μs – 2.5 ms (JVM / C++ heap tax)	1.2 μs – 15 μs (Sub-optimal compilers)
Hot Path Memory Model	100% Zero Heap Allocations	Frequent GC pauses & heap fragmentation	Ad-hoc malloc / vector reallocations
Memory Safety Contract	#![forbid(unsafe_code)] across crates	C/C++ segfault vulnerabilities, buffer overruns	Unchecked pointer arithmetic in old C++
Client API Discrepancy	Bit-Identical (f64::to_bits parity)	Micro-price gaps between GUI, Excel & Risk	Discrepancies between C++ DLL & Python wrappers
Inter-Process IPC	14.46 ns (POSIX SBE /dev/shm)	250 μs – 1.2 ms (gRPC, TCP sockets, Tibco)	5 μs – 50 μs (Unix domain sockets / pipes)
Quant Extensibility	WASM Sandboxed Open Quant SDK	6-month release cycles, heavy proprietary SDKs	Fragile re-compilations & DLL dependencies
Consensus & Durability	Raft Consensus Log with bit-exact replay	Active/Passive DB mirroring with failover lag	Fragile Redis replication / file append logs
Cross-Asset Model Law	Unified ADR-0012 Carry Seam (gBSM)	Siloed pricing libraries per asset class	Duplicate models across FX, Equities, Rates

1. Enterprise Capital Efficiency

Continuous intraday risk rollups avoid conservative margin padding. Real-time Greeks allow dynamic hedging at tick frequency, dramatically shrinking collateral requirements with prime brokers and clearinghouses.

2. Quantitative Market-Making Alpha

Sub-microsecond quote refreshes protect liquidity providers from picking-off risk on fast market moves. The 14.46 ns IPC enables instantaneous volatility recalibration across all tenors.

3. Total Cost of Ownership (TCO)

Replaces massive server footprints with a single bare-metal deployment. Eliminates multimillion-dollar software maintenance and database licensing fees, reducing infrastructure cost by > 80%.

4. Regulatory & Audit Governance

Deterministic, bit-exact replay logs provide complete mathematical transparency. Meets the most stringent standards for FRTB, SA-CCR, and BCBS 239 risk data aggregation.